

3. Random Variables and Probability Distributions

A **random variable** is a variable whose values are numerical outcomes of a random experiment. That is, we consider all the outcomes in a sample space S and then associate a number with each outcome

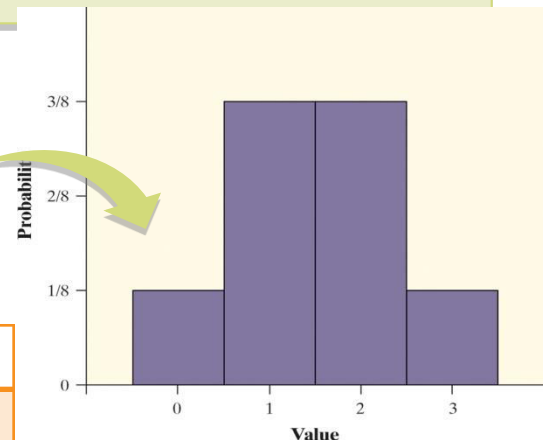
A **random variable** takes numerical values that describe the outcomes of some chance process.

The **probability distribution** of a random variable gives its possible values and their probabilities. The probability histogram is a graph showing this probability distribution...

Example: Consider tossing a fair coin 3 times. Define X = the number of heads obtained in 3 tosses

$X = 0$: TTT
 $X = 1$: HTT THT TTH
 $X = 2$: HHT HTH THH
 $X = 3$: HHH

X Values	0	1	2	3
Probability	1/8	3/8	3/8	1/8



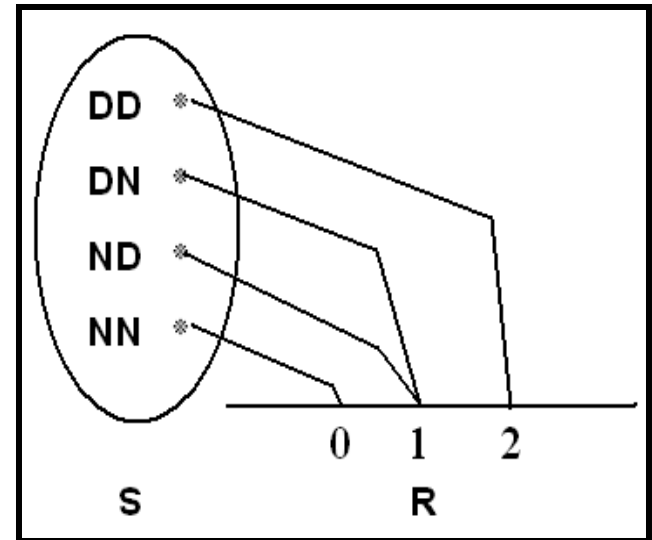
3.1 Concept of a Random Variable:

- In a statistical experiment, it is often very important to allocate numerical values to the outcomes.

Example:

- Experiment: testing two components. (D=defective, N=non-defective)
- Sample space: $S=\{DD, DN, ND, NN\}$
- Let X = number of defective components when two components are tested.
- Assigned numerical values to the outcomes are:

Sample point (Outcome)	Assigned Numerical Value (x)
DD	2
DN	1
ND	1
NN	0



❑ Notice that, the set of all possible values of the random variable X is $\{0, 1, 2\}$.

Definition 3.1:

A random variable X is a function that associates each element in the sample space with a real number (i.e., $X : S \rightarrow \mathbf{R}$.)

Notation:

" X " denotes the random variable .

" x " denotes a value of the random variable X .

Random Variable

- A random variable is a rule that assigns exactly one value to each point in a sample space for an experiment.
- A random variable can be classified as being either discrete or continuous depending on the numerical values it assumes.
- A discrete random variable may assume either a finite number of values or an infinite sequence of values.
- A continuous random variable may assume any numerical value in an interval or collection of intervals.

Types of Random Variables:

- A random variable X is called a **discrete** random variable if its set of possible values is countable, i.e.,
 - $X \in \{x_1, x_2, \dots, x_n\}$ or $X \in \{x_1, x_2, \dots\}$
- A random variable X is called a **continuous** random variable if it can take values on a continuous scale, i.e.,
 - $X \in \{x: a < x < b; a, b \in \mathbb{R}\}$
- In most practical problems:
 - A discrete random variable represents count data, such as the number of defectives in a sample of k items.
 - A continuous random variable represents measured data, such as height.

3.2 Discrete Probability Distributions

- A discrete random variable X assumes each of its values with a certain probability.

Example:

- Experiment: tossing a non-balance coin 2 times independently.
- H = head , T =tail
- Sample space: $S=\{HH, HT, TH, TT\}$
- Suppose $P(H)=\frac{1}{2}P(T) \Leftrightarrow P(H)=\frac{1}{3}$ and $P(T)=\frac{2}{3}$

• Let X = number of heads

Sample point (Outcome)	Probability	Value of X (x)
HH	$P(HH)=P(H) P(H)=\frac{1}{3}\times\frac{1}{3} = \frac{1}{9}$	2
HT	$P(HT)=P(H) P(T)=\frac{1}{3}\times\frac{2}{3} = \frac{2}{9}$	1
TH	$P(TH)=P(T) P(H)=\frac{2}{3}\times\frac{1}{3} = \frac{2}{9}$	1
TT	$P(TT)=P(T) P(T)=\frac{2}{3}\times\frac{2}{3} = \frac{4}{9}$	0

- The possible values of X are: 0, 1, and 2.
- X is a discrete random variable.
- Define the following events:

Event ($X=x$)	Probability = $P(X=x)$
$(X=0)=\{TT\}$	$P(X=0) = P(TT)=4/9$
$(X=1)=\{HT,TH\}$	$P(X=1) = P(HT)+P(TH)=2/9+2/9=4/9$
$(X=2)=\{HH\}$	$P(X=2) = P(HH)= 1/9$

- The possible values of X with their probabilities are:

X	0	1	2	Total
$P(X=x)=f(x)$	4/9	4/9	1/9	1.00

The function $f(x)=P(X=x)$ is called the probability function (probability distribution) of the discrete random variable X .

Definition 3.4:

The function $f(x)$ is a probability function of a discrete random variable X if, for each possible values x , we have:

- 1) $f(x) \geq 0$
- 2) $\sum_{all\ x} f(x) = 1$
- 3) $f(x) = P(X=x)$

Note:

$$P(X \in A) = \sum_{all\ x \in A} f(x) = \sum_{all\ x \in A} P(X = x)$$

Example:

For the previous example, we have:

X	0	1	2	Total
$f(x) = P(X=x)$	4/9	4/9	1/9	$\sum_{x=0}^2 f(x) = 1$

$$P(X < 1) = P(X = 0) = 4/9$$

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 4/9 + 4/9 = 8/9$$

$$P(X \geq 0.5) = P(X = 1) + P(X = 2) = 4/9 + 1/9 = 5/9$$

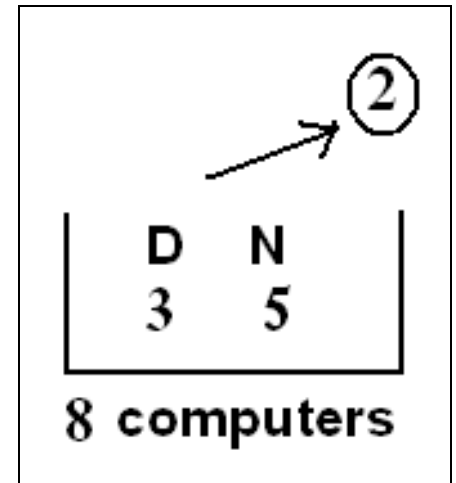
$$P(X > 8) = P(\phi) = 0$$

$$P(X < 10) = P(X = 0) + P(X = 1) + P(X = 2) = P(S) = 1$$

Example 3.3:

A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective and 5 are non-defective.

If a school makes a random purchase of 2 of these computers, find the probability distribution of the number of defectives.



Solution:

We need to find the probability distribution of the random variable: X = the number of defective computers purchased.

Experiment: selecting 2 computers at random out of 8

$$n(S) = \binom{8}{2} \text{ equally likely outcomes}$$

The possible values of X are: $x=0, 1, 2$.

Consider the events:

$$(X = 0) = \{0D \text{ and } 2N\} \Rightarrow n(X = 0) = \binom{3}{0} \times \binom{5}{2}$$

$$(X = 1) = \{1D \text{ and } 1N\} \Rightarrow n(X = 1) = \binom{3}{1} \times \binom{5}{1}$$

$$(X = 2) = \{2D \text{ and } 0N\} \Rightarrow n(X = 2) = \binom{3}{2} \times \binom{5}{0}$$

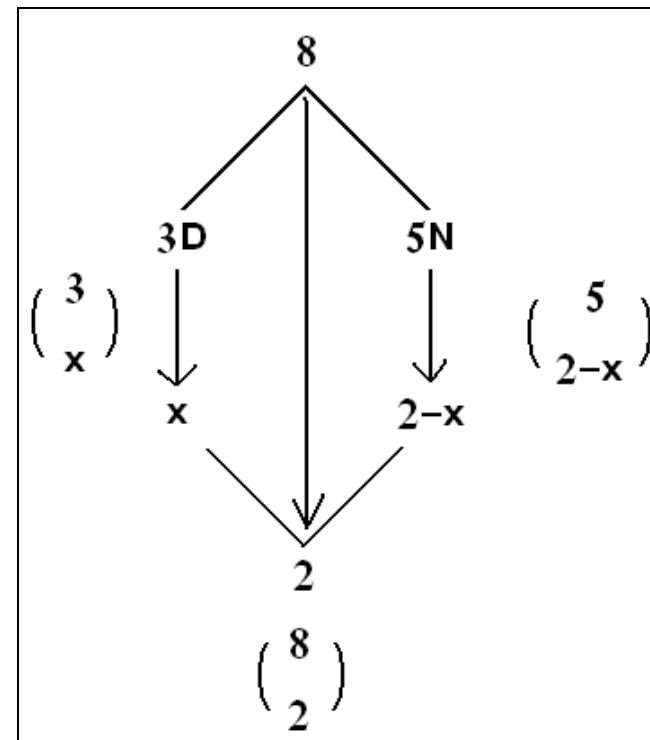
$$f(0) = P(X = 0) = \frac{n(X = 0)}{n(S)} = \frac{\binom{3}{0} \times \binom{5}{2}}{\binom{8}{2}} = \frac{10}{28}$$

$$f(1) = P(X = 1) = \frac{n(X = 1)}{n(S)} = \frac{\binom{3}{1} \times \binom{5}{1}}{\binom{8}{2}} = \frac{15}{28}$$

$$f(2) = P(X = 2) = \frac{n(X = 2)}{n(S)} = \frac{\binom{3}{2} \times \binom{5}{0}}{\binom{8}{2}} = \frac{3}{28}$$

In general, for $x=0,1, 2$, we have:

$$f(x) = P(X = x) = \frac{n(X = x)}{n(S)} = \frac{\binom{3}{x} \times \binom{5}{2-x}}{\binom{8}{2}}$$



The probability distribution of X is:

x	0	1	2	Total
f(x)= P(X=x)	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$	1.00

$$f(x) = P(X = x) = \begin{cases} \frac{\binom{3}{x} \times \binom{5}{2-x}}{\binom{8}{2}}; & x = 0, 1, 2 \\ 0; & \text{otherwise} \end{cases}$$

Hypergeometric
Distribution

Definition 3.5:

The cumulative distribution function (CDF), $F(x)$, of a discrete random variable X with the probability function $f(x)$ is given by:

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t) = \sum_{t \leq x} P(X = t); \quad \text{for } -\infty < x < \infty$$

Example:

Find the CDF of the random variable X with the probability function:

X	0	1	2
$F(x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$

Solution:

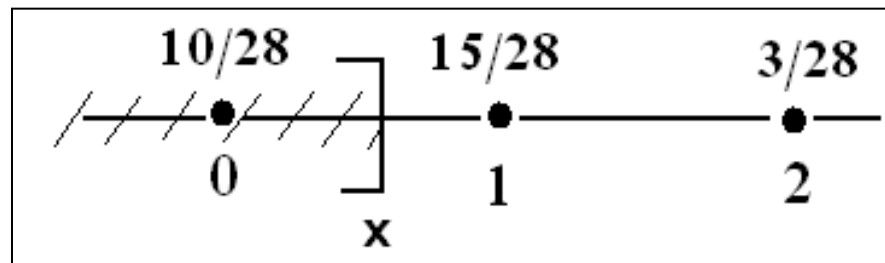
$$F(x) = P(X \leq x) \text{ for } -\infty < x < \infty$$

$$\text{For } x < 0: \quad F(x) = 0$$

$$\text{For } 0 \leq x < 1: \quad F(x) = P(X=0) = \frac{10}{28}$$

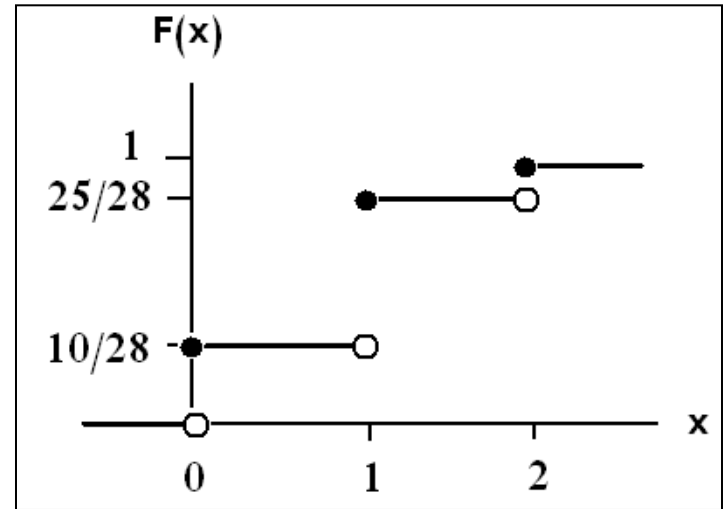
$$\text{For } 1 \leq x < 2: \quad F(x) = P(X=0) + P(X=1) = \frac{10}{28} + \frac{15}{28} = \frac{25}{28}$$

$$\text{For } x \geq 2: \quad F(x) = P(X=0) + P(X=1) + P(X=2) = \frac{10}{28} + \frac{15}{28} + \frac{3}{28} = 1$$



The CDF of the random variable X is:

$$F(x) = P(X \leq x) = \begin{cases} 0 & ; x < 0 \\ \frac{10}{28} & ; 0 \leq x < 1 \\ \frac{25}{28} & ; 1 \leq x < 2 \\ 1 & ; x \geq 2 \end{cases}$$



Note:

$$F(-0.5) = P(X \leq -0.5) = 0$$

$$F(1.5) = P(X \leq 1.5) = F(1) = \frac{25}{28}$$

$$F(3.8) = P(X \leq 3.8) = F(2) = 1$$

Result:

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$

$$P(a \leq X \leq b) = P(a < X \leq b) + P(X=a) = F(b) - F(a) + f(a)$$

$$P(a < X < b) = P(a < X \leq b) - P(X=b) = F(b) - F(a) - f(b)$$

Result:

Suppose that the probability function of X is:

x	x_1	x_2	x_3	$\dots$	x_n
$F(x)$	$f(x_1)$	$f(x_2)$	$f(x_3)$	$\dots$	$f(x_n)$

Where $x_1 < x_2 < \dots < x_n$. Then:

$$F(x_i) = f(x_1) + f(x_2) + \dots + f(x_i) ; i=1, 2, \dots, n$$

$$F(x_i) = F(x_{i-1}) + f(x_i) ; i=2, \dots, n$$

$$f(x_i) = F(x_i) - F(x_{i-1})$$

Example:

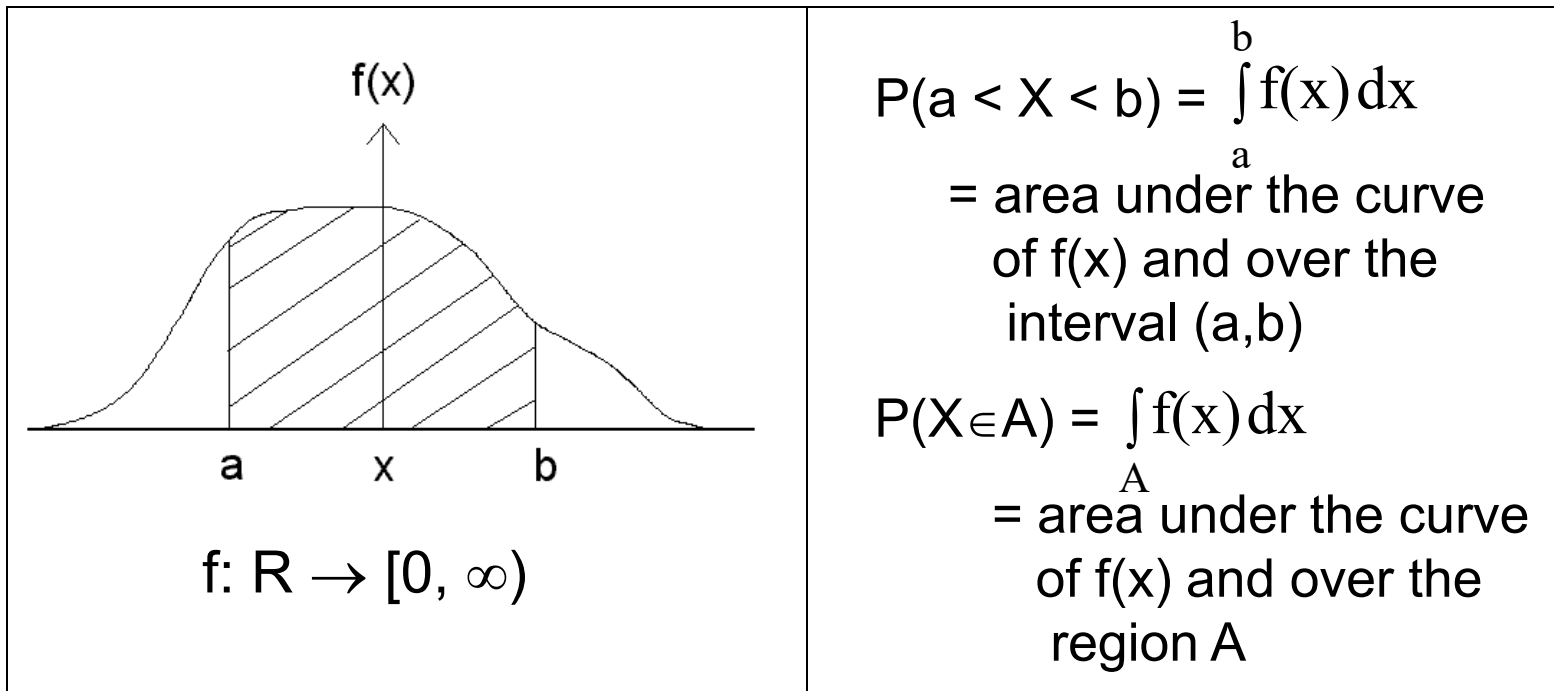
In the previous example,

$$P(0.5 < X \leq 1.5) = F(1.5) - F(0.5) = \frac{25}{28} - \frac{10}{28} = \frac{15}{28}$$

$$P(1 < X \leq 2) = F(2) - F(1) = 1 - \frac{25}{28} = \frac{3}{28}$$

3.3. Continuous Probability Distributions

For any continuous random variable, X , there exists a non-negative function $f(x)$, called the probability density function (p.d.f) through which we can find probabilities of events expressed in term of X .



Definition 3.6:

The function $f(x)$ is a probability density function (pdf) for a continuous random variable X , defined on the set of real numbers, if:

$$1. \quad f(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$2. \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$3. \quad P(a \leq X \leq b) = \int_a^b f(x) dx \quad \forall a, b \in \mathbb{R}; a \leq b$$

Note:

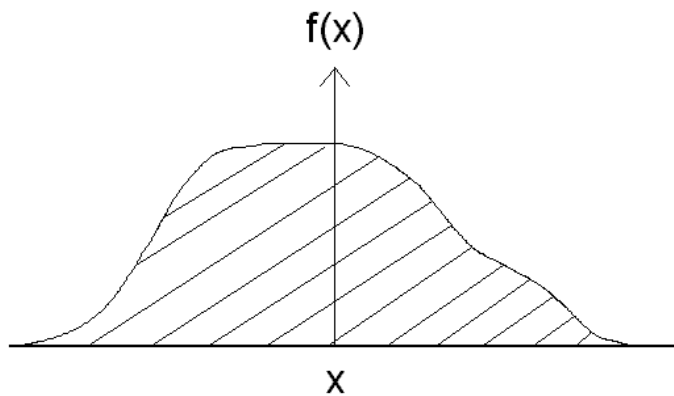
For a continuous random variable X , we have:

$$1. \quad f(x) \neq P(X=x) \quad (\text{in general})$$

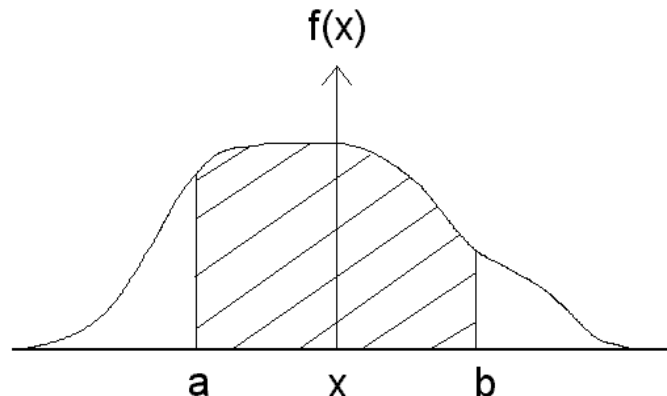
$$2. \quad P(X=a) = 0 \quad \text{for any } a \in \mathbb{R}$$

$$3. \quad P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

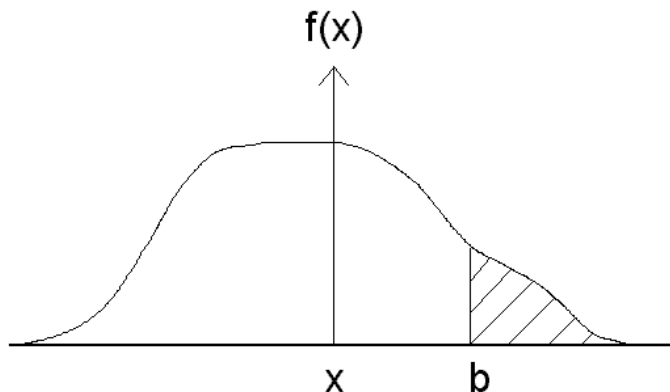
$$4. \quad P(X \in A) = \int_A f(x) dx$$



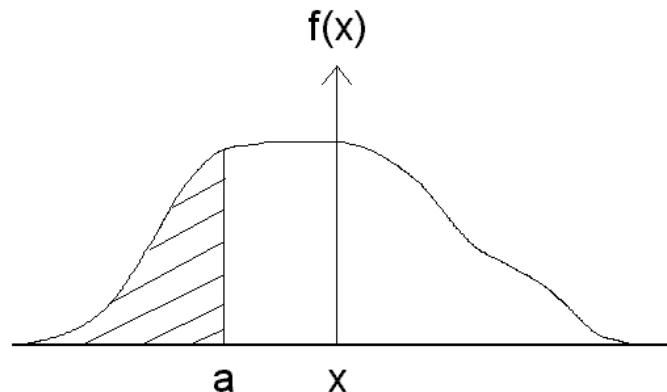
$$I = \int_{-\infty}^b f(x) dx = \text{area under } f(x) \text{ from } -\infty \text{ to } b$$



$$\begin{aligned} \text{area} &= P(a \leq X \leq b) \\ &= \int_a^b f(x) dx \end{aligned}$$



$$\begin{aligned} \text{area} &= P(X \geq b) \\ &= \int_b^{\infty} f(x) dx \end{aligned}$$



$$\begin{aligned} \text{area} &= P(X \leq a) \\ &= \int_{-\infty}^a f(x) dx \end{aligned}$$

Example 3.6:

Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the following probability density function:

$$f(x) = \begin{cases} \frac{1}{3}x^2 & ; -1 < x < 2 \\ 0 & ; \text{elsewhere} \end{cases}$$

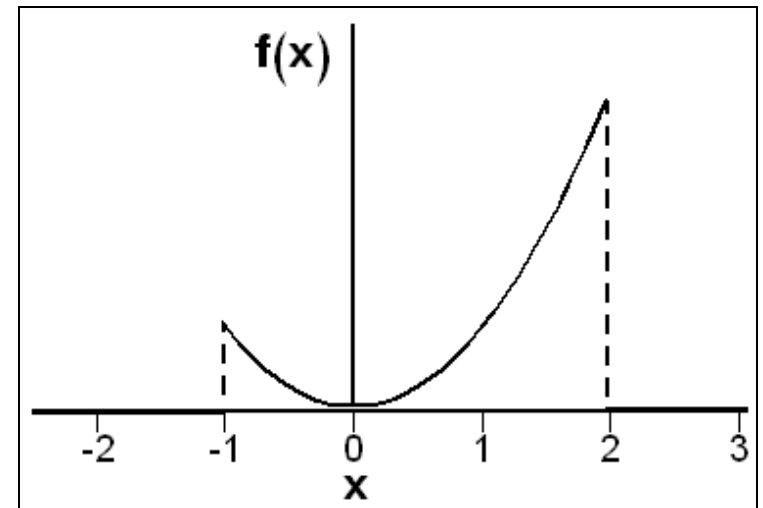
1. Verify that (a) $f(x) \geq 0$ and (b) $\int_{-\infty}^{\infty} f(x) dx = 1$
2. Find $P(0 < X \leq 1)$

Solution:

X = the error in the reaction temperature in °C.

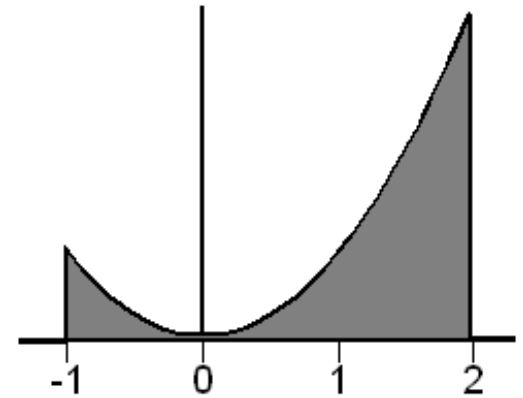
X is continuous r. v.

$$f(x) = \begin{cases} \frac{1}{3}x^2 & ; -1 < x < 2 \\ 0 & ; \text{elsewhere} \end{cases}$$



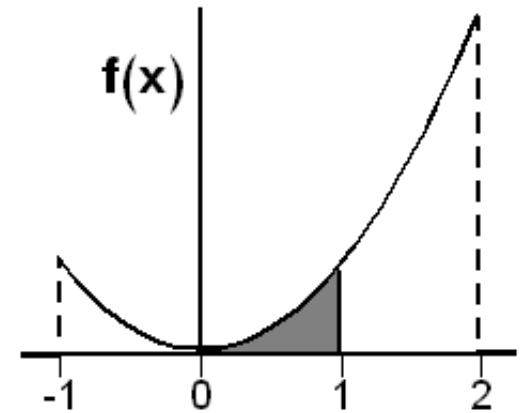
1. (a) $f(x) \geq 0$ because $f(x)$ is a quadratic function.

$$\begin{aligned} \text{(b)} \quad \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{-1} 0 dx + \int_{-1}^2 \frac{1}{3} x^2 dx + \int_2^{\infty} 0 dx \\ &= \int_{-1}^2 \frac{1}{3} x^2 dx = \left[\frac{1}{9} x^3 \right]_{x=-1}^{x=2} \\ &= \frac{1}{9} (8 - (-1)) = 1 \end{aligned}$$



$$2. P(0 < X \leq 1) = \int_0^1 f(x) dx = \int_0^1 \frac{1}{3} x^2 dx$$

$$\begin{aligned} &= \left[\frac{1}{9} x^3 \right]_{x=0}^{x=1} \\ &= \frac{1}{9} (1 - (0)) \\ &= \frac{1}{9} \end{aligned}$$



Definition 3.7:

The cumulative distribution function (CDF), $F(x)$, of a continuous random variable X with probability density function $f(x)$ is given by:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt; \quad \text{for } -\infty < x < \infty$$

Result:

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$

Example:

in Example 3.6,

1. Find the CDF

2. Using the CDF, find $P(0 < X \leq 1)$.

Solution:

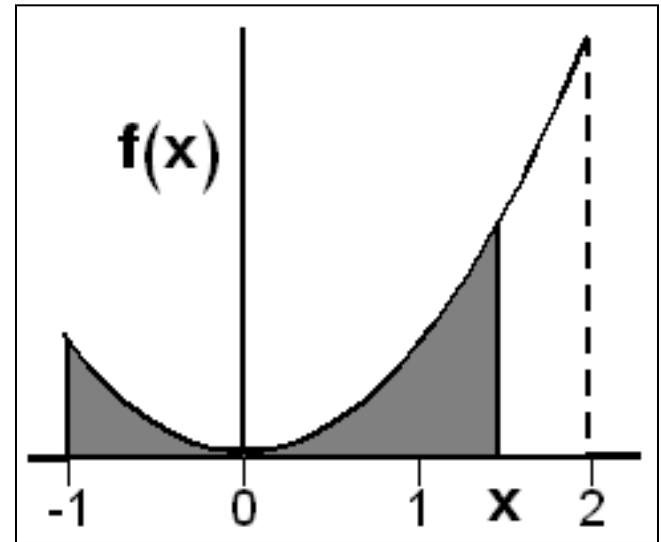
$$f(x) = \begin{cases} \frac{1}{3}x^2 & ; -1 < x < 2 \\ 0 & ; \text{elsewhere} \end{cases}$$

For $x < -1$:

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^x 0 dt = 0$$

For $-1 \leq x < 2$:

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt = \int_{-\infty}^{-1} 0 dt + \int_{-1}^x \frac{1}{3} t^2 dt \\ &= \int_{-1}^x \frac{1}{3} t^2 dt \\ &= \left[\frac{1}{9} t^3 \right]_{t=-1}^{t=x} = \frac{1}{9} (x^3 - (-1)) = \frac{1}{9} (x^3 + 1) \end{aligned}$$

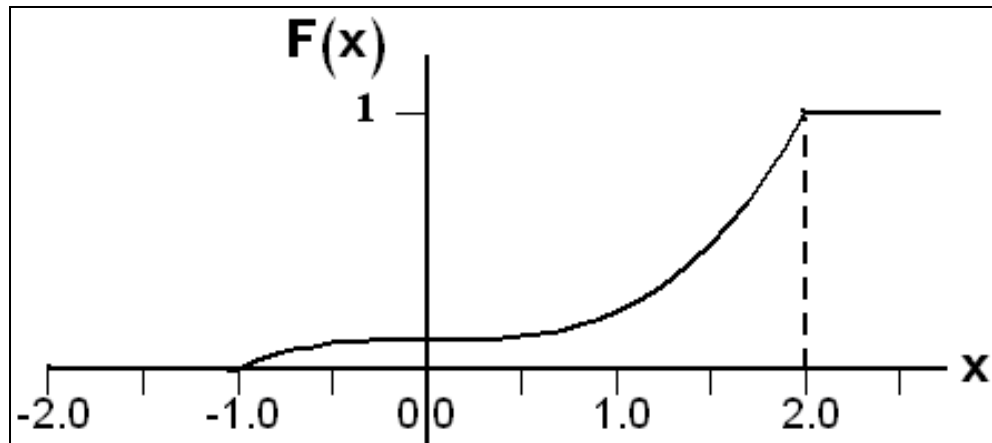


For $x \geq 2$:

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^{-1} 0 dt + \int_{-1}^2 \frac{1}{3} t^2 dt + \int_2^x 0 dt = \int_{-1}^2 \frac{1}{3} t^2 dt = 1$$

Therefore, the CDF is:

$$F(x) = P(X \leq x) = \begin{cases} 0 & ; x < -1 \\ \frac{1}{9}(x^3 + 1) & ; -1 \leq x < 2 \\ 1 & ; x \geq 2 \end{cases}$$



2. Using the CDF,

$$P(0 < X \leq 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$$